MAT 01156
ÁLGEBRA LINEAR II

29 de novembro de 2025

Lista 4 - Soluções

E1

E2 Escreve $v = \langle v, u_1\rangle u_1 + \cdots + \langle v, u_n\rangle u_n$ e $w = \langle w, u_1\rangle u_1 + \cdots + \langle w, u_n\rangle u_n$. Então,

$$\langle v, w\rangle = \left\langle \sum_{i=1}^{n} \langle v, u_i\rangle u_i, \sum_{j=1}^{n} \langle w, u_j\rangle u_j \right\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} \langle v, u_i\rangle \overline{\langle w, u_j\rangle} \langle u_i, u_j\rangle.$$

Como $\{u_1, \ldots, u_n\}$ é ortonormal, $\langle u_i, u_j\rangle = 0$ se $i \neq j$ e $\langle u_i, u_i\rangle = 1$. Portanto,

$$\langle v, w\rangle = \sum_{i=1}^{n} \langle v, u_i\rangle \overline{\langle w, u_i\rangle}.$$

E3

E4

E5

E6

E7

E8

E9

E10

E11